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def vacuum_world():

    # 0 indicates CLEAN, 1 indicates DIRTY
    goal_state = {'A': '0', 'B': '0'}
    cost = 0

    print("NAME: KEERTHANA M")
    print("USN: 1BM24CS135")

    print("\n--- Vacuum Cleaner Environment Setup ---")

    # Get initial location
    location_input = input(
        "Enter initial location of Vacuum (A/B): "
    ).upper().strip()

    # Get status of rooms
    status_input = input(
        "Enter status of Room B (0 for CLEAN, 1 for DIRTY): "
    ).strip()

    other_room = 'B' if location_input == 'A' else 'A'

    status_input_complement = input(
        f"Enter status of Room {other_room} (0 for CLEAN, 1 for DIRTY): "
    ).strip()

    # Create current environment dictionary
    environment = {
        location_input: status_input,
        other_room: status_input_complement
    }

    print(f"\nInitial Environment State: {environment}")
    print("Starting simulation...")

    # Process Room A
    if location_input == 'A':

        # Condition for Room A
        if environment['A'] == '1':
            print("Room A is DIRTY. Sucking the dirt...")
            environment['A'] = '0'
            cost += 1
            print(f"Room A cleaned. Current Cost: {cost}")
        else:
            print("Room A is already CLEAN.")

        # Move to Room B
        print("Moving to Room B...")
        cost += 1
        print(f"Cost + 1 for moving")

        # Condition for Room B
        if environment['B'] == '1':
            print("Room B is DIRTY. Sucking the dirt...")
            environment['B'] = '0'
            cost += 1
            print(f"Room B cleaned. Current Cost: {cost}")
        else:
            print("Room B is already CLEAN.")

    # Process Room B
    else:

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# Condition for Room B
if environment['B'] == '1':
    print("Room B is DIRTY. Sucking the dirt...")
    environment['B'] = '0'
    cost += 1
    print(f"Room B cleaned. Current Cost: {cost}")
else:
    print("Room B is already CLEAN.")

# Move to Room A
print("Moving to Room A...")
cost += 1
print(f"Cost + 1 for moving")

# Condition for Room A
if environment['A'] == '1':
    print("Room A is DIRTY. Sucking the dirt...")
    environment['A'] = '0'
    cost += 1
    print(f"Room A cleaned. Current Cost: {cost}")
else:
    print("Room A is already CLEAN.")

# Final Summary
print("\n--- Final Simulation Results ---")
print(f"Final Environment State: {environment}")
print(f"Goal State Met: {environment == goal_state}")
print(f"Total Path Cost: {cost}")

# Run the program
if __name__ == "__main__":
    vacuum_world()

```

NAME: KEERTHANA M
USN: 1BM24CS135

--- Vacuum Cleaner Environment Setup ---
Enter initial location of Vacuum (A/B): B
Enter status of Room B (0 for CLEAN, 1 for DIRTY): 1
Enter status of Room A (0 for CLEAN, 1 for DIRTY): 1

Initial Environment State: {'B': '1', 'A': '1'}
Starting simulation...
Room B is DIRTY. Sucking the dirt...
Room B cleaned. Current Cost: 1
Moving to Room A...
Cost + 1 for moving
Room A is DIRTY. Sucking the dirt...
Room A cleaned. Current Cost: 3

--- Final Simulation Results ---
Final Environment State: {'B': '0', 'A': '0'}
Goal State Met: True
Total Path Cost: 3

